

Characterization of Visual Information Processing Alterations in Animal Models of Retinal Degeneration

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1. Introduction

Retinal degenerative diseases, including retinitis pigmentosa (RP) and cone-rod dystrophy (CRD), are characterized by progressive photoreceptor loss leading to severe visual impairment and, ultimately, blindness. Although the molecular mechanisms of retinal degeneration are relatively well understood, their impact on central visual processing remains insufficiently characterized, particularly at early stages of disease progression.

2. Methods

In this study, we investigated visual information processing using the RHO^{P23H/WT} rat model of early-stage RP and the GCAP1^{E111V} knock-in mouse model of CRD. All experimental procedures were conducted in accordance with NIH, ARVO, and European guidelines. *In vivo* electrophysiological recordings were performed under isoflurane anesthesia using multi-electrode recordings in the primary visual cortex (V1) and superior colliculus (SC). The local field potentials (LFP) and spiking activity were acquired simultaneously. Visual stimuli were presented on a gamma-corrected display and precisely synchronized with electrophysiological recordings. Visually evoked potentials (VEPs) were computed by averaging responses across repeated trials (100 repetitions per stimulus) to improve signal-to-noise ratio. VEP amplitudes were quantified as peak-to-peak differences, and statistical comparisons were performed using non-parametric tests.

3. Results

In the RHO^{P23H/WT} rat, we observed increased cortical excitability and altered response dynamics at early disease stages, suggesting compensatory or maladaptive changes that precede advanced photoreceptor degeneration. In contrast, GCAP1^{E111V} mice exhibited a progressive reduction in VEP amplitude in both V1 and SC, reflecting a decline in visual signal transmission with disease progression. These effects were consistent across recording sites and supported by quantitative analysis of response amplitudes and variability, revealing distinct, model-dependent patterns of cortical adaptation and dysfunction.

4. Conclusions

These findings demonstrate distinct, model-dependent alterations in central visual processing that emerge early and are likely to influence therapeutic outcomes. Understanding how the visual system reorganizes during disease progression is essential for developing treatments that restore not only retinal structure but also functional vision.

Key words: retinal degeneration, visual processing, electrophysiology, visually evoked potentials

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